

FUNCTIONAL ANALYSIS

(5 hours)

January 11, 2024

Please write on the front page your preferred time for the oral exam.

1. Let l^∞ be the Banach space of bounded sequences with the usual norm.

(i) Show that the linear functional defined for convergent sequences $x = (x_n)$ by

$$l_0(x) = \lim_{n \rightarrow \infty} x_n$$

extends to a continuous linear functional on l^∞ , but such a continuous extension l of l_0 cannot be written in the form

$$l((x_n)) = \sum_{n \geq 1} x_n y_n,$$

for any fixed sequence $(y_n) \in l^1$.

(ii) Show that the set $\mathcal{L}$ consisting of all continuous extensions of l_0 to l^∞ is closed in the weak-star topology on $(l^\infty)'$.

2. Let $M : L^2([0, 1]) \rightarrow C([0, 1])$ be a bounded linear map.

a) Show that M has the form

$$Mf(x) = \int_0^1 K(x, y)f(y)dy, \quad f \in L^2([0, 1]),$$

where $K(x, \cdot) \in L^2([0, 1])$, for every $x \in [0, 1]$, and

$$\sup_{x \in [0, 1]} \|K(x, \cdot)\|_2 < \infty.$$

3. Let X be a Banach space and let $M : X \rightarrow X$ be a linear map. Prove that M is bounded if and only if there exists a set $S \subset X'$, dense in X' , such that for each $l \in S$ the functional m_l defined by

$$m_l(x) = l(Mx), \quad x \in X$$

is continuous on X .

4. Let X, U be Banach spaces, and let $T : X \rightarrow U$ be a compact linear map.

a) Show that the range of T ,

$$R_T = \{Tx : x \in X\},$$

is closed in U if and only if T has finite rank, i.e. $\dim R_T < \infty$.

b) Prove that the range of the transpose T' ,

$$R_{T'} = \{T'l : l \in U'\},$$

is closed in X' , if and only if T has finite rank.

5. Let X be a Banach space and let $T, S : X \rightarrow X$ be bounded linear maps.

(i) Show that if T, S commute ($TS = ST$) then any eigenspace of S ,

$$N_\lambda = \{x \in X : Sx = \lambda x\}$$

is invariant for T , i.e. $TN_\lambda \subset N_\lambda$.

(ii) Prove that if X is a Hilbert space, S is a compact symmetric map and T is a bounded linear map T with the property that any eigenspace of S is invariant for T , then T commutes with S .

6. Let H be a Hilbert space and let $A, B \in \mathcal{L}(H)$ be injective compact symmetric linear maps such that:

(i) $\sigma(A) = \sigma(B)$,

(ii) All eigenspaces of A and B have dimension 2.

Prove that there exists a bijective isometry (unitary map) $U \in \mathcal{L}(H)$ such that $A = UBU^{-1}$. Also show by means of an example that the conclusion may fail if we assume only the hypothesis (i).